Enhancing Nailfold Capillary Segmentation Through Improved U-Net Network Architecture

Adekunle Timothy ADEWOYE
Medical Imaging Analysis
M.Sc. Computer Vision and Robotics, VIBOT
University of Burgundy
Le Creusot, 71200
adewoyetimothy@gmail.com

Supervising Professor - Mahya Faraji Zamharir

***Abstract*— Nailfold capillaroscopy (NC) is a useful non-invasive diagnostic method that doctors frequently employ to evaluate the microcirculation in a patient's capillaries. It is especially crucial to assess the images produced by NC when diagnosing conditions where there is a change in the morphology of the capillaries. Unfortunately, the majority of NC photos are of low quality, making analysis challenging and time-consuming. Therefore, our work aimed to find a method for accurately segmenting the capillaries in low-quality NC pictures. Our suggestion was to use a deep neural network with a Res-Unet topology to achieve this. To create an encoding-decoding network and deepen the layers inside the network in order to preserve the properties of the deep layer, the network combines the residual network (ResNet) and the U-Net.**

I. INTRODUCTION

The term "microcirculation" describes the blood flow in arterioles. According to Koscielny et al. (1998), human tissue cells can absorb nutrients, take in oxygen, and expel metabolites through capillaries and venules (Tian and Li (1992). Clinicians use capillaroscopy, a straightforward and non-invasive technique that can be used to diagnose and identify autoimmune diseases that affect the microcirculation, such as systemic sclerosis (SSc), Raynaud's phenomenon (RP), and other rheumatologic diseases (Karbalaie, 2018; Mannarino et al., 1994; Cutolo and Smith, 2013). Nailfold capillaroscopy (NC) is a useful method since it is frequently used on capillaries in the nailfold area, where it is easy to examine them (Nivedha et al., 2016).

A hairpin structure is the characteristic physiological pattern of the capillary loop in the nailfold area. The convexity of the capillary head is a crucial factor in differentiating between aberrant and normal capillary morphology (Smith et al., 2016; Cutolo et al., 2018; Karbalaie et al., 2019). Additional morphological characteristics include the density of the capillary loops and the capillaries' diameter, shape, and dispersion (Karbalaie et al., 2019; Faggioli et al., 2015). Capillaries can become dilated and convoluted and capillary microhemorrhages can occur in situations such heart disease, diabetes, and hypertension (Bonacci et al., 1996). For instance, individuals with Raynaud's phenomenon often have an aberrant capillary pattern, such as a changed configuration of capillary loops, a reduction in the quantity of

Even while NC has demonstrated potential in diagnosis, the corresponding low-quality photos, which come from inexpensive cameras, low-contrast images,

Air bubbles in the immersion oil, dust on the lenses, patient fingers moving during imaging, and image noise all significantly impede the capacity of doctors to measure and segment the capillaries and distinguish between linked disorders (Nivedha et al., 2016; Shigeharu et al., 2016; Li et al., 2017). This paper presents a method to precisely segment capillaries on low-quality NC pictures, thereby mitigating the impact of these images.

A few methods for autonomously extracting capillaries have been suggested in earlier research. Using an enhancement operation, for instance, N. P. Doshi et al. pre-processed nailfold capillary images before skeletonizing them. They found that an iterative rule-based skeletonization process combined with a straightforward α-trimmed filter worked best to address non-uniform lighting (Doshi et al., 2015). Tama et al. (2015) conducted a study in which they processed nailfold capillaroscopy images in four stages: pre-processing, binarization, skeleton extraction, and skeleton segmentation. Following binarization, skeleton extraction, and skeleton segmentation, morphological characteristics could be measured.

The capillary profiles in both of the aforementioned investigations were successfully segmented or the capillary skeletons were extracted using the methods employed, yet the NC pictures in those

Although the methods utilized in the two aforementioned research were successful in extracting the capillary skeletons or segmenting the capillary profiles, the NC images used in both investigations were of a good quality and had minimal background noise. However, segmenting capillaries in low-quality NC pictures is crucial in practice. Using advances in artificial intelligence could be one method of interpreting low-quality NC photos. Particularly with regard to CNN (convolutional neural networks) (Haoxiang et al., 2015; Sombro et al., 2017) and networks that have been further improved, like FCN (fully convolutional network) (Shelhamer et al., 2017) and U-Net, deep learning has already demonstrated good performance in medical image processing (Shen et al., 2017; Lawrence Carin et al., 2018).

notably with regard to CNN (convolutional neural networks) (Haoxiang et al., 2015; Sombro et al., 2017) and further improved networks like FCN (fully convolutional network) (Shelhamer et al., 2017) and U-Net, which provide pixel-level classification and can generate a prediction for each pixel while preserving the spatial information of the original input image. As a result, these networks may be able to significantly lessen the issues related to image quality in order to guarantee the consistency and repeatability of capillary segmentation outcomes.

In order to separate capillaries from background, this research suggests a capillary segmentation technique based on Res-Unet. The FCN network provided the U-shaped structure, which makes it possible to obtain additional feature maps from lower layers. Three sections summarise the paper: Section one covers the image acquisition and pre-processing procedure; Section 2 describes the segmentation algorithms applied in the paper; and Section 3 concludes that Res-Unet performs well in the segmentation of nailfold capillary images based on qualitative and quantitative comparison tests.

Materials

Nailfold capillary data collection

The number of fingers or fields that can be chosen for testing is unlimited; in this case, the fourth finger was examined (Karbalaie, 2018). Prior to the examination, patients were allowed to settle in and become comfortable for fifteen to twenty minutes at room temperature (25 to 30 °C) (Etehad Tavakol et al., 2015). Next, the hand of the patient—the one that was not as injured—was positioned at the level of the heart (Etehad Tavakol et al., 2015).

To improve transparency and lessen diffuse reflection, a small amount of cedar oil was applied to the skin near the nail's base. We employed a non-contact microcirculation microscope (model XW880) equipped with a white LED light source and a 5 million-pixel CMOS camera for imaging.

The apparatus operated at a magnification of 380×. The device's fixator was touched with the fourth finger, and the focusing knob was turned.

To get the best view of each finger's capillary image, a CCD camera was mounted and connected to a computer.

Thirty individuals between the ages of twenty and fifty made up the study's participant pool: five had hypotension, three had hypertension, and fifty were healthy. The training set consisted of 30 cases, and 20 were used as the test set.

Pre-processing

The gathered picture data required to be pre-processed in order to build the training and testing set for the U-Net network. Initially, meticulously trimmed the area of interest on every picture, ensuring that every cropped image had an identical resolution of 512 × 512. The obtained RGB capillary images were low contrast, whereas images in the green channel are supposed to exhibit a strong contrast between the capillaries and the backdrop, according to literature (AI-Gindya et al., 2008). Thus, in order to compare for greater contrast, a grayscale is obtained and green channel capillary pictures.

Although there was no discernible change in the contrast between the two images, it was discovered that the grayscale images were linked to superior performance; that being said, the green channel images might be more illuminating (Nivedha et al., 2016).

Data Augumentation

There were about thirty-four photos in the training dataset. some of the resultant label images were improved using the ImageDataGenerator function in Keras for horizontal flipping, translation, amplification, rotation, and shear. This increases the network's robustness and invariance; vertical flipping was avoided because it would alter the image's semantic information in the object hierarchies (Li, 2011). The training image's pixel value matrix was placed in the first channel during augmentation, and the corresponding pixel value matrix of the labelled image was placed in the third channel. The channels were then combined into a single image for data augmentation. This allowed the labelled image and the training image to change simultaneously. Following that, the labeled image and the training image separated according to the channels. After augmentation, the number of images in the training set was 20664

II. RELATED WORKS

Previous studies have proposed various methods for autonomously extracting capillaries from nailfold capillaroscopy images. Doshi et al. (2015) introduced an iterative skeletonization process combined with a filtering technique to address non-uniform lighting conditions. Tama et al. (2015) developed a segmentation pipeline involving preprocessing, binarization, and skeleton extraction stages for morphological analysis. While these methods were effective in segmenting capillary structures, they were evaluated using high-quality images and may not generalize well to low-quality datasets.

Recent advancements in artificial intelligence, particularly convolutional neural networks (CNNs), have shown promise in medical image processing tasks (Haoxiang et al., 2015; Sombro et al., 2017). Networks like FCN and U-Net, which enable pixel-level classification and preserve spatial information, offer potential solutions to address image quality issues in capillary segmentation (Shelhamer et al., 2017; Shen et al., 2017). The Res-Unet architecture, combining features of both ResNet and U-Net, presents a promising approach for accurately segmenting capillaries from low-quality images

III. METHODOLOGIES

U-Net

U-Net's longconnected jumping structure and U-shaped structure are its key characteristics. According to Ronnberger et al. (2015), U-Net is an encoder-decoder network with an elongated U form that is composed of two paths: a contraction path and an expansion path. The path of contraction is a convolutions network architecture that consists of two repeated 3 × 3 convolutions and ReLu activation functions, followed by a 2 × 2 max pooling. After each max pooling, the number of feature channels is doubled. The path of expansion and contraction are symmetrical.

ReLu activation functions, two 3 × 3 convolutions, and a 2 × 2 up sampling make up each layer (Ronneberger et al., 2015). Both low-level information and high-level semantics can be concatenated using the long-connected jumping structure. A 1

ResNet

Feature extraction is greatly influenced by the network's depth and classification, since a shallow network has less capacity for recognition (Zhang et al., 2019; Szegedy et al., 2015; Simonyan and Zisserman, 2014).

capacities. Deeper layers are required to extract more details from capillaries with low contrast. Theoretically, a network functions better the deeper it is. However, in actuality, as a network gets deeper, the training set's accuracy declines because the gradient vanishes and the network responds to training less well (Bengio et al., 1994; Glorot and Bengio, 2010). A residual network (ResNet) can be used to tackle this

issue (Szegedy et al., 2016).

ResNet offers two possible solutions to the issue of training accuracy declining with network depth: identity mapping and residual mapping. The network will only leave identity mapping when the weights of its numerous nonlinear layers of residual mapping are driven toward zero, having achieved the ideal balance between depth and accuracy. Therefore, the performance of the network will not decrease with an increase in depth once it achieves its theoretically ideal state (He et al., 2016a).

Based on the original unit structure, as seen in Fig. 3(b), a new residual unit structure with improved performance is proposed (He et al., 2016a; He et al., 2016b). For x = x + F(x), the result is H(x) = 1 when there are 1 layers. Assuming that it begins at layer 1, the final layer L's output is as follows:

$$H(xL) = x + \sum_{i=1}^{L} F(xi)$$

As illustrated in Fig. 3(c), the characteristic H(xL) can be written as the sum of x and the residual function F(x) between each layer. The gradient will not vanish, as the formula's reverse derivation makes clear.

Res-Unet

We suggest a new deep neural network called Res-Unet in light of the benefits of ResNet and U-Net (Ronneberger et al., 2015; He et al., 2016a). Higher-level features are extracted using ResNet as the front-end feature extraction module, which is similar to down sampling (use ResNet18 as an example) (He et al., 2016b). Fig. 4 displays the Res-Unet neural network.

Three parts make up the Res-Unet: coding, decoding, and lengthy connection. Feature extraction, which involves downsampling to retrieve high-pixel information from context, occurs throughout the coding phase. Subsequently, the decoding section resumes with the piXels classification. The purpose of the lengthy connecting section is to combine the features that were taken from the coding and decoding sections.

Two leftover blocks are added to each layer during the coding phase. Identity mapping and two 3 × 3 convolution blocks are present in every residual block. Various ResNets, including ResNet18, ResNet34, and so on, are composed of varying amounts of residual blocks (Szegedy et al., 2015). The coding and decoding halves of the structure are symmetric, forming a U-shaped structure with four levels in the decoding component that each contain two residual units. A feature map from the matching layer of the contraction path is fused with each layer. The details are guaranteed by the jumping structure. Before each layer, the underlying feature map is up-sampled to guarantee the same re-solution in fusion. Sigmoid and a 1 × 1 convolution make up the final layer.

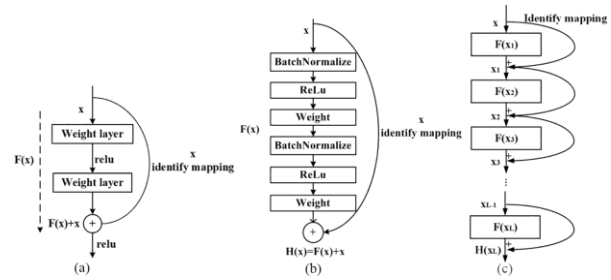


Fig. 1. Residual block. (a) original residual block (He et al., 2016a). The dashed line is the residual mapping. (b) proposed residual block. The curve is identity mapping, indicating the easiest paths for the information to propagate (He et al., 2016b). (c) the multi-layer network with residual blocks.

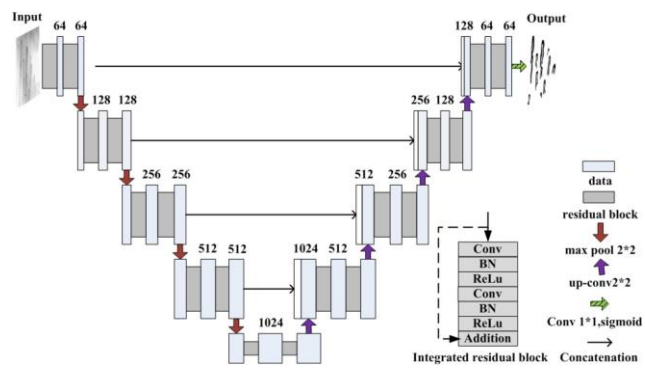


Fig. 2. Res-Unet18 is a neural network structure with a gray-scale input and binary output, multi-channel feature maps, concatenated copied maps, convolution layers, BN layers, ReLu layers, and an integrated residual block.

Accuracy of segmentation results.

	Median	Mean	SD
U-Net	86.90	86.24	1.74
Res-Unet18	91.32	91.72	1.39
Green channel	90.79	90.81	1.26
Res-Unet34	91.41	91.41	1.57

Dice scores of segmentation results.

	Median	Mean	SD
U-Net	97.40	97.29	5.92
Res-Unet18	97.59	97.66	5.42
Green channel	97.28	97.34	5.83
Res-Unet34	97.23	97.39	6.14

Green channel: For this model, input images of Res-Unet18 were in the green channel; the input images of other models were in gray scale. Accuracy (%): (TP + TN)/(TP + TN + FP + FN); TP: true positive, actually positive, pre- dicted positive; TN: true negative, actually negative, predicted negative; FP: false positive, actually negative, predicted positive; FN: false negative, actually positive, predicted negative; SD: standard deviation. Dice (%): 2TP/(2TP + FN + FP).

IV. RESULTS AND ANALYSIS

Numerous comparative tests was carried out to find a compromise between training complexity and network accuracy. We present our findings regarding the optimal network architecture here. One among the test which was carried out is shown in fig. 3, 4 and 5. Indicating the Comparison of precision recall curves and ROC curves for the models, the accuracy rate of the models and loss rate of the models.

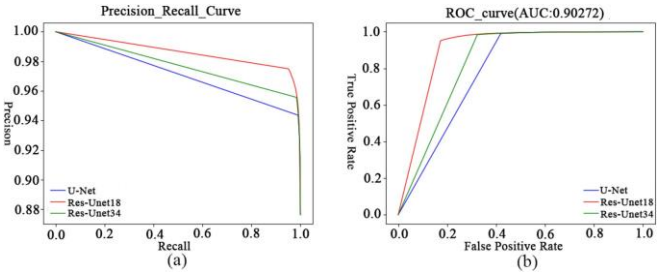


Fig. 3. Comparison of precision recall curves and ROC curves for the models. (a) Precision recall curves. Res-Unet18 has the highest precision rate or highest recall rate. (b) ROC curves. True Positive Rate (TPR): the rate of correctly predicted positive classes in all positive classes; False Positive Rate (FPR): the rate of predicted positive in all negative classes. The AUC value of the Res-Unet18 model is 0.9027 and convex to the upper left, meaning that the segmentation accuracy of Res- Unet18 was the highest.

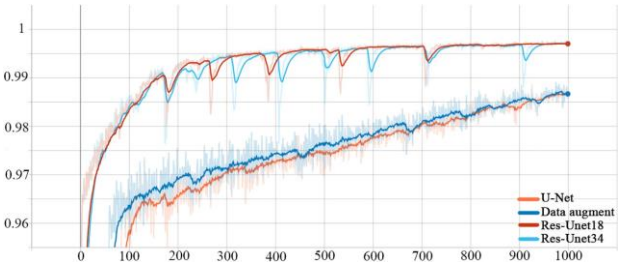


Fig. 4. The accuracy rate of models.

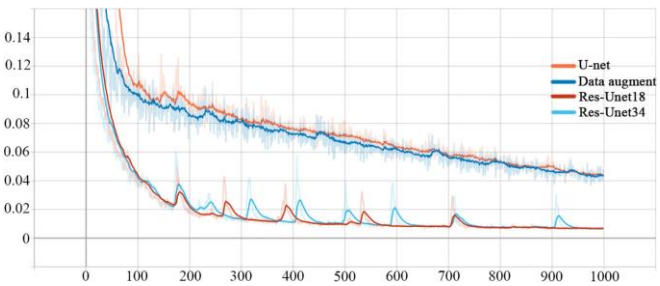


Fig. 5. The loss rate of models.

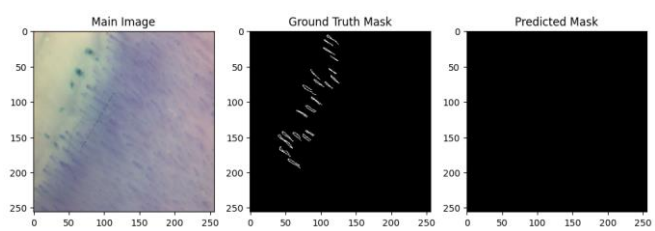
Quantitative Evaluation

For the models of U-Net, a quantitative evaluation on 34 test images with 34 marks was carried out. that show the results of training the three models regarding accuracy and dice scores, main image, ground truth mask and predicted mask. As shown below

To qualitatively evaluate the model's performance, an example image along with its ground truth mask and predicted mask were visualized:

- **Main Image:** The original nailfold capillaroscopy image.
- **Ground Truth Mask:** The manually annotated mask indicating the regions of capillaries in the image.
- **Predicted Mask:** The mask predicted by the trained U-Net model.

The qualitative evaluation provides insights into how well the model is able to segment capillaries in unseen images. By comparing the predicted mask with the ground truth mask, we can assess the accuracy and efficacy of the segmentation.



$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{TPR} = \frac{TP}{TP + FN}$$

$$\text{FPR} = \frac{FP}{FP + TN}$$

Figures 4 and 5 display the networks' training process curves.
The accuracy or loss rate is represented by the ordinate, while the abscissa shows the number of rounds in a network's 100 iterations. Out of the three models, Res-Unet18 exhibited the highest training accuracy and the fastest-declining loss rate with the lowest numerical value. After 560 rounds, the accuracy rate was 0.996, a significant improvement over the original U-Net. The convergence was smooth. This suggests that segmentation can be done effectively using the residual network in conjunction with U-Net.
We also looked at the impact of enhancing the U-Net model with additional data. Based on the graphs, the model with data augmentation had a higher training accuracy than the original model.

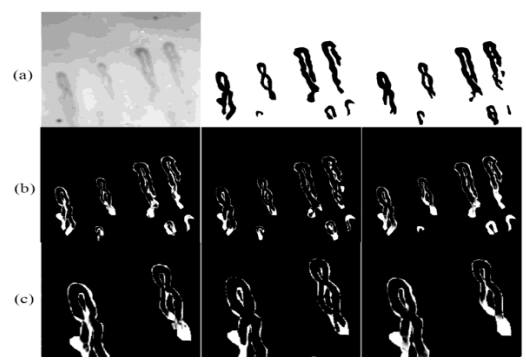


Fig. 6. Comparison of the segmentation results of crossed capillaries. (a) input, label, and segmentation result of Res-Unet18. (b) differences between label andpredictions of U-Net, Res-Unet18, and Res-Unet34 (different area is in white). (c) the details of differences.

V. RESULTS

The U-Net model was trained on a dataset consisting of 49 training images and their corresponding masks, with an additional 21 images reserved for testing. After training for 3 epochs, the model achieved the following results:

Training Loss: Starting from an initial loss of approximately 1.68, the loss decreased consistently over the epochs to reach 1.63.

Validation Loss: The validation loss followed a similar trend, starting from 1.66 and decreasing to 1.61 by the end of training.
These quantitative metrics indicate a gradual improvement in the model's performance in terms of minimizing the loss function, suggesting that the model was effectively learning to segment nailfold capillaries from the images.

A Comparison of the segmentation results of crossed capillaries with differences between label and predictions of U-Net, Res-Unet18, and Res-Unet34 is shown in the figures below.

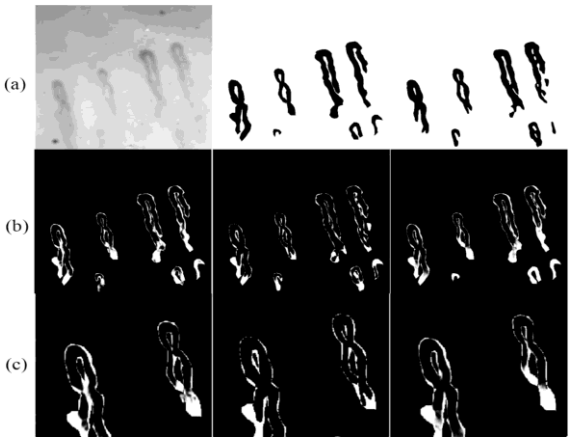


Fig. 8. Comparison of the segmentation results of crossed capillaries. (a) input, label, and segmentation result of Res-Unet18. (b) differences between label and predictions of U-Net, Res-Unet18, and Res-Unet34 (different area is in white). (c) the details of differences.

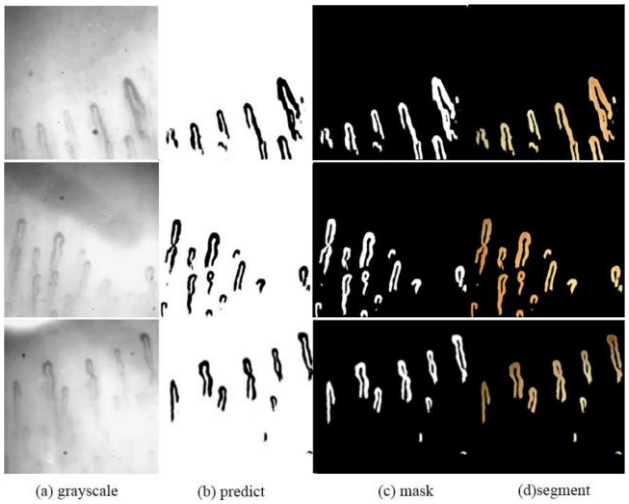


Fig. 9. Segmentations of Res-Unet18. (a) the gray-scale input images. (b) the predictions of the Res-Unet18 network. (c) mask: the inverse operation of binarizationof predictions. (d) the segmentation after mask operation.

VI. CONCLUSION

Nailfold capillaroscopy can be used to detect changes in the microvasculature in patients, and microangiopathy is associated with significant clinical problems (Ghizzoni et al., 2015). Indeed, research has found that certain traits of the capillaries in the nailfold are linked to particular illnesses; these traits include capillary dilation (CD), capillary tortuosity (CT), and capillary enlargement (CE) (Pancar & Kaynar, 2018). Determining the clinical condition of a disease therefore requires the ability to quantify and evaluate the appearance of capillaries in the nailfold.

This study suggested using a neural network that combines U-Net and ResNet to improve the often poor quality of nailfold capillary images. More capillary details were preserved in low-quality photos thanks to this network's leaping connection, coding-decoding structure, and deeper layers.

In addition, the results demonstrate the potential of the U-Net architecture in enhancing nailfold capillary segmentation. The model showed improvements in both quantitative loss metrics and qualitative segmentation accuracy, indicating its effectiveness in addressing the challenges associated with low-quality nailfold capillaroscopy images. These findings support the hypothesis that deep learning-based approaches, such as the U-Net model, hold promise for improving the diagnosis and management of autoimmune diseases affecting microcirculation. Further validation on larger datasets and clinical trials would be necessary to assess the generalizability and real-world utility of the proposed method.

However, out of the three models that were examined, the Res-Unet18 model performed the best in both assessments and tests. Following the

After the segmentation image was created, the morphological traits of the nailfold capillaries could be easily measured and calculated. Thus, even when only poor-quality images are available, this method can be used in clinical auXiliary judgement of related diseases.

REFERENCES

AI-Gindya, A., AI-Ahmadb, H., QahwajiC, R., Tawfika, A., 2008. A novel blind image water-marking technique for color rgb images in the dct domain using green channel. In: Mosharaka International Conference on Communications, Computers and Applications, pp. 26–31.

Bengio, Y., Simard, P., Frasconi, P., 1994. Learning long-term dependencies with gradient descent is difficult. IEEE Trans. Neural Netw. 5 (2), 157–166.

E. Bonacci, N. Santacroce, N. D'Amico, R. Mattace. “Nail-fold capillaroscopy in the study of microcirculation in elderly hypertensive patients,” Arch. Gerontol. Geriatr. Suppl. 5, vol. 22, pp. 79–83, 1996, doi:https://doi.org/10.1016/0167-4943(96)86918-4.

Bourquard, A., Pablo-Trinidad, A., Butterworth, I., et al., 2018. Non-invasive detection of severe neutropenia in chemotherapy patients by optical imaging of nailfold microcirculation. Sci. Rep. 8 (5301).

Cutolo, M., Smith, V., 2013. State of the art on nailfold capillaroscopy: a reliable diagnostic tool and putative biomarker in rheumatology? Rheumatology 52 (11), 1933–1940.

Cutolo, M., Melsens, K., Herrick, A.L., Foeldvari, I., Deschepper, E., De Keyser, F., Distler, O., Ingegnoli, F., Mostmans, Y., Müller-Ladner, U., 2018. Reliability of simple capillaroscopic definitions in describing capillary morphology in rheumatic diseases. Rheumatology 57 (4), 757–759. https://doi.org/10.1093/rheumatology/kex460.

Doshi, N.P., Schaefer, G., Zhu, S.Y., 2015. An evaluation of image enhancement techniques for nailfold capillary skeletonisation. Procedia Comput. Sci. 60, 1613–1621.

Etehad Tavakol, M., Fatemi, A., Karbalaie, A., Emrani, Z., Erlandsson, B.-E., 2015. Nailfold capillaroscopy in rheumatic diseases: which parameters should be evaluated? Biomed. Res. Int. 1–17. https://doi.org/10.1155/2015/974530.

Faggioli, P., Tamburello, A., Sciascera, A., Gilardi, A.G., Mazzone, A., 2015. Nailfold video capillaroscopy in internal medicine. Italian J. Med. 9 (3), 234–242.

Ghizzoni, Cecilia, Sebastiani, Marco, Manfredi, Andreina, Campomori, Federica, Colaci, Michele, Giuggioli, Dilia, Ferri, Clodoveo, 2015. Prevalence and evolution of scleroderma pattern at nailfold videocapillaroscopy in systemic sclerosis patients: clinical and prog- nostic implications. Microvasc. Res. 99, 92–95.

Glorot, X., Bengio, Y., 2010. Understanding the difficulty of training deep feedforward neural networks. J. Mach. Learn. Res. 9, 249–256.

HaoXiang, Li, Lin, Zhe, Shen, Xiaohui, et al., 2015. A convolutional neural network cascade for face detection. In: IEEE Conference on Computer Vision and Pattern Recognition, pp. 5325–5334.

He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition. In: 2016a IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 770–778.

He, K., Zhang, X., Ren, S., Sun, J., 2016b. Identity mappings in deep residual networks. In: Computer Vision and Pattern Recognition, . https://arXiv.org/abs/1603.05027.

Ioffe, S., Szegedy, C., 2015. Batch normalization: accelerating deep network training by redu-cing internal covariate shift. In: Machine Learning, . https://arXiv.org/abs/1502.03167.

Karbalaie, Abdolamir, 2018. Novel analysis toolkit for capillaroscopic images. In: Development and Clinical Evaluation.

Karbalaie, Abdolamir, Emrani, Zahra, Fatemi, Alimohammad, Etehadtavakol, Mahnaz, Erlandsson, Björn-Erik, 2019. Practical issues in assessing nailfold capillaroscopic images: a summary. Clin. Rheumatol. 38 (9), 2343–2354.

Khan1, Asifullah, Sohail1, Anabia, Zahoora1, Umme, Qureshi1, Aqsa Saeed, 2019. A survey of the recent architectures of deep convolutional neural networks. In: Computer Vision and Pattern Recognition, . https://arXiv. org/abs/1901.06032.